ALGEBRAIC POINTS ON CURVES

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Lecture 1

[STARTED LATE]

Example 1. $C = \mathbb{P}^1$, then

$$\mathbb{P}^1(\overline{K}) = \mathbb{A}^1(\overline{K}) = \overline{K} \cup \{\infty\},$$

Alternatively

$$\mathbb{P}^1(\overline{K}) = \{\infty\} \cup \bigcup_{d \in \mathbb{N}} \bigcup_{\substack{f \in k[x] \\ \deg(f)=d \\ \text{irred}}} \left\{\alpha \in \overline{K} \ : \ f(\alpha) = 0\right\},$$

i.e. we can partition closed points by their degree. Degree d points on $\mathbb{P}^1_{\overline{K}}$ are parametrised by homogeneous irreducible degree d polynomials in $K[x_0, x_1]$ modulo scaling. These are then contained in the set of homogeneous degree d polynomials, which is parametrised by $\mathbb{P}^d(K)$. In fact this is equal to the rational points on the associated Hilbert scheme

$$\left\{\text{degree } d, \text{ 0 dimensional } K\text{-subschemes of } \mathbb{P}^1\right\} =: \mathrm{Hilb}^d_{\mathbb{P}^1}(K).$$

We are interested in $C(\overline{K})$ and the action of G_K, equivalently closed points on C, and we have partitioned these by the degree d.

$$C(\overline{K}) \leftrightarrow \{\text{closed points on } C\} = \bigsqcup_{d \in \mathbb{N}} \{\text{degree } d \text{ points on } C\}.$$

We will then embed the right hand side similarly to the above in $\mathrm{Hilb}^d_C(K)$.

Question 2. *Can we understand **all** closed points of C?...or perhaps a **Zariski dense** set of closed points?*

Note that the following are not Zariski dense

- $C(K)$,
- The set of points with a fixed residue field, $\left\{x \in C(\overline{K}) \ : \ k(x) = L\right\}$.

Example 3. $C : y^2 = -2(x^2-2)(x^2-3)(x^2-2x-4)\dots h(x)$ has a Zariski dense set of *quadratic points.*

Question 4. *Our example with quadratic points used the degree 2 map $C \to \mathbb{P}^1$ induced by $(x, y) \mapsto y$. What if $C \to \mathbb{P}^1$ has degree $d > 2$?*

In this case we can use Hilbert's irreduciblity theorem to show that if we have a degree d map then the degree d points are Zariski dense.

Hence the following are Zariski dense

- The set of closed points,
- the set of quadratic points (i.e. points of degree 2), if $C : y^2 = f(x)$,
- the set of degree d points when there exists a degree d map $C \to \mathbb{P}^1$

Definition 5. The density degree set is

$$\delta(C/K) := \left\{ d \in \mathbb{N} \ : \ {}^{\text{degree } d \text{ points}}_{\text{Zariski dense on } C} \right\}.$$

We have

$$\delta(C/K) \supseteq \left\{ \deg(C \to \mathbb{P}^1) \ : \ C \to \mathbb{P}^1 \text{ nonconstant} \right\},$$

and we call the right hand side the *Lüroth semigroup* $\delta_{\mathbb{P}^1}(C/K)$

Note that

$$\{\text{degree } d \text{ points on } C\} \subseteq \{\text{degree } d \text{ 0-dimensional subschemes of } C\} = \mathrm{Hilb}^d_C = \mathrm{Sym}^d_C.$$

If $d \in \delta(C/K)$ then there exists a positive dimensional $Z \subseteq \mathrm{Sym}^d_C$ with Zariski dense K-points. Indeed, since $\{x \in C \ : \ \deg(x) = d\}$ is Zariski closed hence infinite, and this set embeds as a subset $Z' \subseteq \mathrm{Hilb}^d_C(K)$ and so must contain a positive dimensional Z since infinite sets of points are not Zariski closed.

Question 6. *What are the positive dimensional $Z \subseteq \mathrm{Sym}^d_C$ with Zariski dense K-points?*

Well we can also think of the degree d 0-dimensional subschemes of C as degree d divisors on C. Hence we can map to the divisor class, and obtain a map to Pic^d_C which thus induces a natural map $\rho : \mathrm{Sym}^d_C \to \mathrm{Pic}^d_C$.

Assume $\dim(\rho(Z)) = 0$. Then $Z \subseteq |D| \cong \mathbb{P}^N$ for some $N \geq 1$. Note that $\mathbb{P}^N$ contains at least one $\mathbb{P}^1$! When you translate the definitions, we can then use this to obtain a degree d map to $\mathbb{P}^1$, recovering the first construction that we saw. Hence $Z = |D|$ give $\mathbb{P}^1$-parametrized points. We shall show how to construct this map tomorrow.

Remark 7. This $\mathbb{P}^1$ gives us two divisors D_0, D_1 which are linearly equivalent, i.e. $D_0 - D_1 = \mathrm{Div}(f)$ and f will induce our map.

Definition 8. A closed point $x \in C$ is $\mathbb{P}^1$-parametrized if any (equivalently all) of the following hold.

- $\exists \pi : C \to \mathbb{P}^1$ with $\pi(x) \in \mathbb{P}^1(K)$ and $\deg(\pi) = \deg(x)$;
- $\exists \mathbb{P}^1 \hookrightarrow \mathrm{Sym}^d_C$ whose image contains x;
- $h^0(X, \mathcal{O}(x)) \geq 2$, where we interpret x as a divisor.

Otherwise we say that x is $\mathbb{P}^1$-isolated.

Now assume that $\dim(\rho(Z)) > 0$. Since $\rho(Z)$ has Zariski dense K-points, and is contained in Pic^d_C which is (pretty much) an abelian variety. This is one of the few situations we understand! We can appeal to Mordell–Lang (proved by Faltings) which shows then that $\rho(Z)$ is a positive rank abelian variety. Such Z give so-called AV-parametrised points.

Definition 9. A closed point $x \in C$ is AV-parametrised if there is a positive rank abelian subvariety $B \subset \mathrm{Pic}^0_C$ such that $[x] + B \subseteq W^d$, where $W^d := \rho(\mathrm{Sym}^d_C)$. We write $\delta_{\mathrm{AV}}(C/K)$ for the set of degrees for which there is a AV-parametrised point of that degree. Otherwise we say that x is AV-isolated.

Definition 10. We will say that a point is parametrized if it is either AV or $\mathbb{P}^1$ parametrized. It is isolated if it is neither.

We have a corollary of Faltings 1994 proof of Mordell–Lang.

Theorem 11 (Bourdon–Ejder–Liu–Odumodu–Viray, 2019)**.** *Let C/K be a nice curve.*

(1) $\delta(C/K) = \delta_{\mathbb{P}^1}(C/K) \cup \delta_{\mathrm{AV}}(C/K)$;
(2) There are finitely many isolated points on C. In other words, there is an open set $U \subseteq C$ such that every closed $x \in U$ is parametrized.

Question 12. *Can we understand all parametrized points on C?*

Lecture 2

Today we will look at characterising $\mathbb{P}^1$-parametrised points, with a view towards proving Theorem 11.

Theorem 13 (Faltings 1994, Mordell–Lang Conjecture)**.** *Let A be an abelian variety over K and let $X \subseteq A$ be a closed subvariety. Then there exist $x_1, \dots, x_r \in X(K)$ and $B_1, \dots, B_r \subseteq A$ abelian subvarieties such that*

(1) $x_i + B_i \subseteq X$ for all i, and
(2) $X(K) = \bigcup_{i=1}^r x_i + B_i(K)$.

Note that if there is $[D] \in \mathrm{Pic}^d_C(K)$ then we have a K-isomorphism $\mathrm{Pic}^d_C \cong \mathrm{Pic}^0_C$ by translation by $[D]$. For all intents and purposes, since we are only interested in $d \in \delta(C/K)$, we can always assume that $\mathrm{Pic}^d_C(K) \neq \emptyset$ and so we may assume that Pic^d_C is an abelian variety.

We apply Theorem 13 to $A = \mathrm{Pic}^0_C \cong \mathrm{Pic}^d_C \supseteq W^d = X$. Thus there exist $[D_1], \dots, [D_r] \in W^d(K)$ and $B_1, \dots, B_r \subseteq \mathrm{Pic}^0_C$ such that

$$[D_i] + B_i \subseteq W^d$$

and

$$W^d(K) = \bigcup_i [D_i] + B_i(K).$$

$$\{\text{isolated points on } C\} \subseteq \{\text{AV-isolated points}\}$$

and the right hand side maps through ρ_d into $W^d(K)$ and so is contained in $\cup_i [D_i] + B_i(K)$ and moreover we may only union over the B_i with rank 0 as otherwise such points would be AV-parametrised. Hence, by the Mordell–Weil theorem, this union is a finite set and so we have finitely many AV-isolated points.

Assume that $d \in \delta(C/K)$. Then $\{\text{degree } d \text{ points on } C\} \subseteq \mathrm{Hilb}^d_C(K)$ is infinite. Mapping down under ρ_d we get infinite image or there are infinitely many points in a single fibre of ρ_d.

In the former case, if $\rho_d(\{\text{degree } d \text{ points on } C\}) \subseteq W^d(K)$ is infinite then there exists $x \in C$ of degree d such that $[x] \in [D_i] + B_i$ with B_i of positive rank. Hence there exists an AC-parametrised degree d point.

In the latter case, we take the equivalent (to be shown!) definition of AV-parametrised points that there exists a $\mathbb{P}^1$ parametrised point.

Hence we have shown $\delta(C/K) \subseteq \delta_{\mathbb{P}^1}(C/K) \cup \delta_{\mathrm{AV}}(C/K)$, modulo:

(1) If there exists a degree d parametrised point then $d \in \delta(C/K)$ + Defineorem;
(2) (a) $\rho_d|_{\{\text{isolated points}\}}$ is finite-to-one;
    (b) If $d \gg 0$ then there are no degree d isolated points.

Lemma 14 (Defineorem)**.** *Take a closed point $x \in C$ of degree $d = \deg(x)$. Then the following are equivalent*

(1) x is $\mathbb{P}^1$-parametrized;
(2) there exists a non-constant morphism $\mathbb{P}^1 \to \mathrm{Hilb}^d_C$ whose image contains x;
(3) $h^0([x]) \geq 2$

Proof. Starting at the first constraint, note that by definition there is a map $\pi : C \to \mathbb{P}^1$ with $\pi(x) \in \mathbb{P}^1(K)$ of degree d. Hence the pullback

$$\pi^* : \mathbb{P}^1 \to \mathrm{Hilb}^d_C$$

satisfies $\infty \mapsto \pi^*(\infty) = [x]$. Hence $1 \implies 2$.

Now starting at the second constraint, the compisition

$$\mathbb{P}^1 \to \mathrm{Hilb}^d_C \xrightarrow{\rho_d} \mathrm{Pic}^d_C$$

must map to a singleton point because abelian varieties contain no rational curves! Hence everything maps to $[x]$. Then there exist $D_1 \neq D_2$ effective divisors such that $[D_1] = [x] + [D_2]$ and there exists $f_i \in K(C)$ such that $\mathrm{Div}(f_i) = D_i - x$. Consider the span $\langle f_1, f_2 \rangle \subseteq H^0(C, \mathcal{O}([x]))$, then this is 2-dimensional by the definition of the D_i and so $2 \implies 3$.

Finally, let us start at the third constraint. Then there exists $f \in K(C)^\times$ nonconstant such that $\mathrm{Div}(f) + [x] \geq 0$. Then $f : C \to \mathbb{P}^1$ is a morphism such that $x \mapsto \infty$. Moreover x is integral (as a scheme!) and maps to a degree 1 point, and so $\deg(f) = \deg(x) = d$. Hence $3 \implies 1$. □

Lemma 15. *Fix $d \in \mathbb{Z}_{>0}$. The following are equivalent*

(1') $\exists \pi : C \to \mathbb{P}^1$ of degree d;
(2') $\exists D \in \mathrm{Div}(C)$ of degree d such that $h^0([D]) \geq 2$ and $|D|$ is basepoint-free;
(3') there exists a degree d $\mathbb{P}^1$-parametrised point.

Sketch proof. (3') $\implies$ (2') follows from Lemma 14 plus ε. For (2') $\implies$ (1') we use the same argument as in Lemma 14 for (3) $\implies$ (1) but use the basepoint-free condition to get $\deg(f) = d$. It remains to show (1') $\implies$ (3').

Given a degree d map $\pi : C \to \mathbb{P}^1$, how do we determine if the set of points $\{z \in \mathbb{P}^1(K) \ : \ C_z \text{ is integral}\}$ is nonempty? If it is, then such a fibre will immediately give us a degree d $\mathbb{P}^1$-parametrised point. Assume that $\pi : C \to \mathbb{P}^1$ is étale over some x and a Galois map with Galois group G. Then the fibre above z will break up into some sum of points $x_1, \ldots, x_r$, and because Galois permutes these by automorphism we obtain $\deg(x_i) = \deg(x_j)$ for all i, j. Moreover, the stabilisers $\mathrm{Stab}(x_i)$ are all conjugate. Write $G_z := \mathrm{Stab}(x_1) \leq G$. Then C_z is integral if and only if $G_z = G$.

If $G_z \subseteq H \subsetneq G$ is contained in a proper maximal subgroup then we have a diagram

$$\begin{tikzcd} C \arrow[dr] \arrow[dd] & \\ & C/H \arrow[dl] \\ \mathbb{P}^1 & \end{tikzcd}$$

and $z \in \operatorname{im}\big((C/H)(K) \to \mathbb{P}^1(K)\big)$, which is a degree > 1 map. Note that this image is a type II thin set (recall that type I thin sets are proper Zariski closed subsets). Hence, taking the finite union over all of these H, we have sketched that $\exists \Omega \subseteq \mathbb{P}^1(K)$ a thin set such that $\forall z \in \mathbb{P}^1(K) \backslash \Omega$, C_z is integral. Then we conclude by Hilberts irreducibility theorem that there exist z outside of this set and so the claim holds. $\square$

We used Hilbert, so we should recall it.

Theorem 16. *Let $\Omega \subseteq \mathbb{P}^1(K)$ be a thin set, then $\mathbb{P}^1(K) \backslash \Omega$ is Zariski dense in $\mathbb{P}^1(K)$.*

In fact we can add another constraint:

(4') there exist infinitely many degree d $\mathbb{P}^1$-parametrized points.